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RESEARCH ARTICLE

Bridging gaps in daily global ocean-colour products using physical fields and neural networks

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Abstract

Satellite ocean-colour observations provide key information on phytoplankton variability and marine biogeochemistry, but their use is strongly constrained by extensive data gaps caused by cloud cover and irregular sampling. Because missingness is highly structured in space and time, it can bias climatologies and trend analyses. Here, we test whether adding ocean–atmosphere surface fields improves deep-learning gap filling of daily global ocean-colour products at 1° resolution. We train convolutional neural networks to reconstruct missing values in three widely used datasets of chlorophyll-a (Chl-a) and phytoplankton size classes (PSC) spanning 1998-2023. To inform reconstructions where ocean-colour observations are sparse, we include predictors from the GLORYS12 ocean reanalysis and ERA5 atmospheric reanalysis, describing surface temperature, salinity, winds, mixing conditions, and incoming light. Including physical predictors consistently improves Chl-a reconstructions across all datasets, reducing reconstruction errors in withheld (masked) pixels, with the largest gains in regions where ocean-colour sampling is most intermittent. Improvements for PSC are smaller and more dataset-dependent, highlighting the influence of product-specific uncertainties and extreme sparsity. Overall, this approach enables the generation of more temporally complete and sensor-consistent ocean-colour fields, supporting more robust analyses of variability and long-term trends.

Impact Statement

Adding surface physical information improves deep-learning gap filling of daily global ocean-colour fields. The resulting reconstructions reduce temporal sampling artefacts from clouds and irregular satellite coverage, enabling more robust analyses of long-term variability and trends. Performance is competitive with state-of-the-art methods, and physical predictors consistently enhance chlorophyll-a reconstruction skill.

1. Introduction

Ocean colour remote-sensing datasets are derived from satellite measurements of seawater reflectances (Ćatipović et al. [2023]). They bring information on photosynthetic organisms living in the upper layers of the ocean, phytoplankton. These play a crucial role in marine ecosystems and biogeochemical cycles (Sarmiento and Gruber [2006],Falkowski [2012]). Remote-sensed ocean colour measurements provide a proxy for phytoplankton biomass through chlorophyll-a (Chl-a), and additional pigments can be used to infer information about phytoplankton community structure (El Hourany et al. [2024],Xi

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et al. [2020],Sathyendranath et al. [2019],Yu et al. [2023]). Continuous satellite measurements of ocean colour have been available since late 1997, beginning with SeaWiFS (Hooker), resulting in more than 25 years of ocean colour time series. However, the number of satellites observing the surface ocean colour to estimate Chl-a and other pigments concentrations varies through time (see Fig 1a). Multiple sensors and orbital differences lead to sampling biases in pigment datasets (van Oostende et al. [2022]). In pigment datasets, the percentage of missing values over time varies with the launch and end of satellite missions, as shown in Figure 1 for three widely used Chl-a datasets (Garnesson et al. [2019],Pauthenet et al. [2024]). One way to address these biases is through the gap filling of pigment time series.

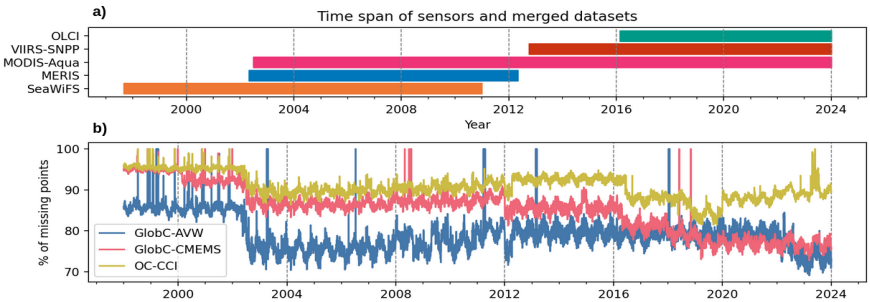

Figure 1. *Ocean colour sensor timeline and percentage of missing values over time for different Chl-a products.*

Deep-learning provides a powerful framework to address this challenge [Zhang et al. [2025]]. Numerous studies have applied deep-learning approaches to gap filling in satellite ocean colour data (Ćatipović et al. [2023], Hong et al. [2023], Barth et al. [2022], Li et al. [2023], Ehrler and Ernst [2021]). We focus on daily fields at coarse (1°) resolution to target large-scale, basin-to-global patterns rather than fine or mesoscale features, while keeping the problem computationally tractable. Although many studies aggregate ocean-colour data to weekly or monthly means to reduce missingness, such aggregation can still leave gaps and may introduce sampling biases when cloud cover and observation availability vary in space and time. In addition, temporal averaging smooths short-lived events and phenological transitions, reducing the ability to analyse day-to-day variability and decreasing the number of training samples available for learning. For these reasons, we perform completion on daily 1° fields at the global scale. To mitigate the impact of sparse observations, we incorporate in our method ocean surface physical fields building on concepts already introduced by previous similar studies (Roussillon et al., Barth et al. [2022], Hong et al. [2023], Martinez et al. [2020]). Indeed, the physical state of the ocean plays a key role in phytoplankton phenology and distribution (Bopp et al. [2013], Dutkiewicz et al. [2013], Messié and Chavez [2012], Lewandowska et al. [2014]). This motivates the following question:

Does the inclusion of physical surface fields improve deep-learning-based gap filling of daily global ocean colour data at coarse spatial resolution?

In this study, we develop a deep-learning approach to fill missing values in global ocean colour data at 1° spatial resolution, with the goal of producing complete daily fields enabling comparative trend analyses. To this end we selected three differents datasets of ocean colour and a reanalysis dataset of ocean surface physics and atmospheric data to train on. Then we compare the completion performance with state-of-the-art architecture and see if adding physics allow a better gap filling in ocean colour data.

2. Method

2.1. Data

2.1.1. Ocean colour data

Chl-a is a widely used product as proxy for phytoplankton biomass, derived from remote-sensing reflectances (O'Reilly and Werdell [2019],Finenko et al. [2003]). In addition, to better characterize phytoplankton community structure, several studies have derived pigment-based indicators from ocean colour observations, as specific pigments act as markers for different phytoplankton groups. In this study, we focus on a size-based classification of phytoplankton communities derived from these pigment concentrations (El Hourany et al. [2024],Xi et al. [2020],Vidussi et al. [2001],Uitz et al. [2006]). Three phytoplankton size classes are considered: microphytoplankton (20–200 µm), nanophytoplankton (2–20 µm), and picophytoplankton (< 2 µm). These size classes are are expressed in terms of relative contribution to the total Chla concentration, hereafter referred to as Phytoplankton Size Classes (PSC). We selected three widely used Chl-a datasets associated with PSC products (Pauthenet et al. [2024]):

- The Ocean colour Climate Change Initiative v6.0 (OC-CCI) Sathyendranath et al. [2019]. The associated PSC data are computed from the same Rrs band using an approach developped in Brewin et al. [2015].
- A GlobColour product, the GlobColour AVW (GlobC-AVW). The associated PSC are computed using El Hourany et al. [2019].
- Antoher Globcolour product, GlobColour CMEMS (GlobC-CMEMS) Garnesson et al. [2019] . The PSC are derived from Xi et al. [2020].

Each dataset spans from 1998 to 2023 originally found at 4km daily resolution.

Dataset	Global missing: Chl-a (%)	PSC (%)	+-50° latitude missing: Chl-a (%)	PSC (%)
OC-CCI	86.15	86.41	80.03	80.36
GlobC-CMEMS	77.80	93.08	68.40	90.31
GlobC-AVW	68.93	82.60	55.77	74.88

Table 1. *Percentage of missing values for Chl-a and PSC in the different datasets for latitudes below 50° and globally*

Table 1 highlights the scarcity of valid observations in the daily fields: missing values can reach 93% for the GLOBC-CMEMS PSC product.

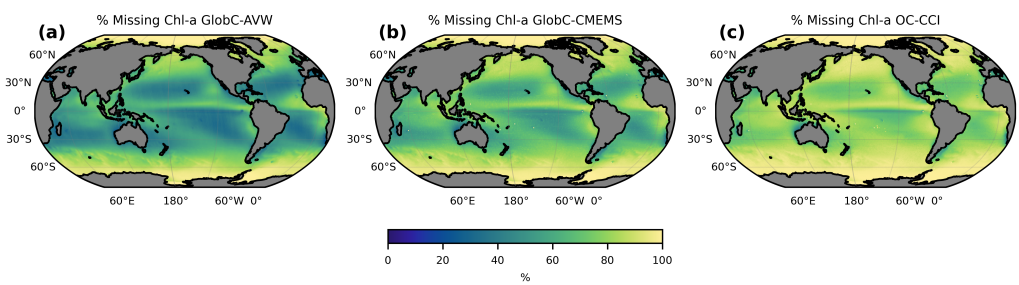

Figure 2. *Spatial distribution of missing Chlorophyll-a (Chl-a) observations (percentage of missing daily values) in the three datasets: (a) GlobC-AVW, (b) OC-CCI, and (c) GlobC-CMEMS. Values are computed on 1° grid over the full record, highlighting strong spatial heterogeneity in observation availability.*

Furthermore, since these missing values are caused by cloud cover or gaps in satellite coverage, they are spatially heterogeneous, as illustrated in Figure 2 for the three Chl-a products. Gyres and the

open ocean at latitudes below 50° contain the most observed areas, except in regions with persistent cloud cover, such as the Equatorial Pacific and Atlantic. Then, the number of missing pixels increases toward higher latitudes, reaching almost 100% above 60° (partially explains by the permanent night time during winter at the poles). While some studies exclude high-latitude regions (Martinez et al. [2020], Roussillon et al.) and others focus specifically on them (Serra-Pompei et al. [2024]), we retain these areas to preserve a spatially continuous global domain (e.g., Southern Ocean and subpolar basins). We note, however, that because observations are extremely limited in these regions, the reconstruction there is weakly constrained by ocean-colour targets and should be interpreted with caution.

2.1.2. Ocean physics data

Phytoplankton variability is strongly influenced by the physical state of the ocean (Bopp et al. [2013]). To allow the reconstruction of missing ocean colour observations, we therefore incorporate surface, gap-free, physical drivers and evaluate their contribution to gap-filling performance.

We use daily sea surface variables at 1/12° resolution from the GLORYS12 global ocean reanalysis over the period 1998–2024 (E.U. Copernicus Marine Service Information (CMEMS) [2023]). These include sea surface temperature (SST) and salinity (SSS), which influence phytoplankton metabolism and stratification, as well as sea surface height (SSH), mixed-layer depth (MLD), and the norm of zonal and meridional surface currents to represent large-scale circulation and mixing processes. Sea Surface Ice were added as well. Melting ice reduces light availability while supplying nutrient-rich freshwater that can influence phytoplankton growth (Naëck et al. [2025]).

In addition, we incorporate daily atmospheric fields from the ERA5 reanalysis at 25 km resolution (Copernicus Climate Change Service (C3S) [2023]), including 10 m wind components and clear-sky shortwave radiation. The wind components primarily influence mixing processes, while the shortwave radiation provides a consistent proxy for the baseline light regime driving phytoplankton growth (Behrenfeld et al. [2006], Frouin and Chertock [1992]).

2.2. Architecture

We implement a traditionnal convolutional autoencoder, illustrated in Figure3 following the SmaAt-UNet architecture proposed by Trebing et al. Trebing et al. [2021]. It replaces the standard convolution with depthwise separable convolution (DSC Chollet [2017], Howard et al. [2017]). DSC applies a spatial convolution independently to each input channel, followed by a 1×1 pointwise convolution that combines the resulting features across channels. By decoupling spatial and channel-wise correlations, DSCs significantly reduce the number of network parameters (from ≈ 20 million to ≈ 4 million parameters)). Convolutional Block Attention Modules (CBAM Woo et al. [2018]) are added before the skip connection. They enhance relevant features in the encoder by learning channel and spatial attention, which are then propagated to the decoder. By decoupling channel and spatial attention, CBAM improve performance and robustness to noise while keeping computational and parameter overhead low, making it easy to integrate into CNN architectures such as U-Net.

The overall structure is identical to the original architecture, with a small number of adaptations to better suit oceanographic data. Specifically, circular padding is applied to the lateral boundaries of the convolutional layers to account for the sphericity of Earth fields. The network output is generated using a 1×1 convolutional layer. The first three output channels are passed through a softmax activation to estimate the PSC% while the fourth channel corresponds to the completed Chl-a field.

A neural network is trained to complete each pigment dataset independantly. The architecture with physics inputs is referred to as SmaAt_phy (see Table 2).

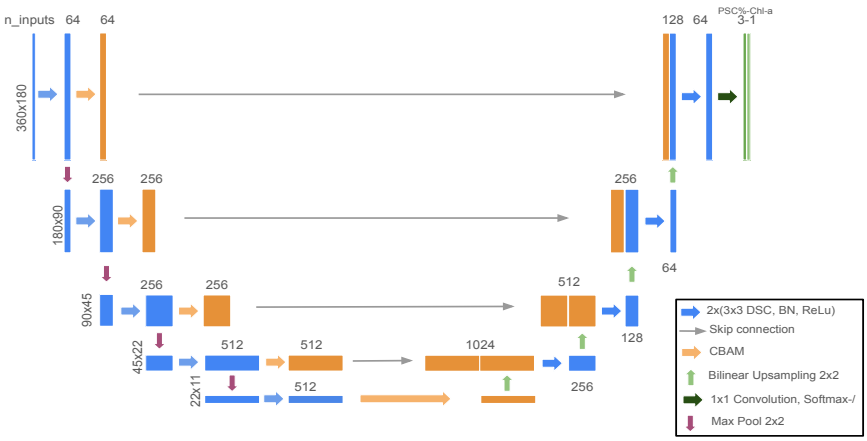

Figure 3. *SMaAt-Unet architecture scheme.*

2.3. Preprocessing

All datasets are regridded to a daily 1° spatial resolution. To enable supervised learning and independent testing, we generate synthetic cloud-like masks by randomly sampling missing-data masks from the ocean-colour products. These masks mimic realistic cloud-induced gaps while providing ground truth for evaluation on withheld pixels. Missing values in the input fields are set to zero (alternative strategies were tested but did not significantly change the results).

The three datasets are randomly splitted into training, validation and testing set. Training is composed of 80% of the images (shape 7596 days × 360 longitude × 180 latitude), validation to test between each epoch is composed of 10% of the images (shape 950 days × 360 longitude × 180 latitude) as is the test set (shape 950 days × 360 longitude × 180 latitude). Randomly splitting timesteps allows the network to learn spatial gap-filling while preserving the intrinsic temporal variability of the datasets. Since the objective is to reconstruct spatially complete fields rather than to predict temporal evolution, this strategy prevents the model from learning artificial temporal dependencies while ensuring exposure to the full range of seasonal and interannual variability present in the data. All physical input fields are standardized by subtracting their mean and dividing by their standard deviation both computed from the training set. The phytoplankton size class (PSC) variables are expressed as relative contributions between 0 and 1, providing a well-bounded and consistent input range for the neural network. Chl-a concentrations are first clipped to the range 10^{-5}–10 mg m^{-3}, to get rid of potential unrealistic oceanic values, then log-transformed and standardized.

2.4. Loss function

The loss function is a Mean Squared Error (MSE) on the masked pixels:

$$L(\hat{Y}, Y) = \frac{\sum_{i \in M} (y_i - \hat{y}_i)^2}{n} \tag{1}$$

where $\hat{Y}$ is the output (Chl-a or PSC), Y is the corresponding target, M is the mask, and n is the number of masked pixels.

2.5. Evaluation

To evaluate our results, standard metrics are computed between output and target on test set. For the Chl-a, the root mean squared error (RMSE) is computed:

$$RMSE(\hat{Y}, Y) = \sqrt{\frac{\sum_{i \in M}(y_i - \hat{y}_i)^2}{n}} \tag{2}$$

where $\hat{Y}$ is the output Chl-a, Y is the target Chl-a, M is the mask, and n is the number of masked pixels.

Spatial reconstruction skill for Chl-a is evaluated using a normalized root mean square error (NRMSE) computed at each grid point.

$$NRMSE(\hat{Y}, Y) = \sqrt{\frac{\sum_{i \in M}(y_i - \hat{y}_i)^2}{\sum_{i \in M}(y_i)^2}} \tag{3}$$

For the PSC we compute a soft extention of the Intersection over Union (IOU). Indeed PSC are percentages and the error magnitude is less intersting than the relative composition of the PSC and the agreement (overlap) between target PSC and completed PSC Rezatofighi et al. [2019].

$$IOU(P\hat{S}C, PSC) = \frac{1}{3} \sum_{c \in psc} \frac{\sum_i \min(x_{i,c}, \hat{x}_{i,c})}{\sum_i \max(x_{i,c}, \hat{x}_{i,c})} \tag{4}$$

To evaluate our approach, we conduct two comparative experiments on OC-CCI, GlobC-CMEMS and GlobC-AVW , using the same training, validation, and test splits. First, we train an identical neural network architecture using only Chl-a and PSC data, excluding physical drivers, in order to assess the added value of physical information referred to as SmaAt. Second, we compare the quality of the reconstructed Chl-a fields with those produced by DINCAE, an autoencoder developped in Barth et al. [2020] for ocean data completion and widely used in the domain (Barth et al. [2022],Han et al. [2020],Ji et al. [2021],Jung et al. [2022],Luo et al. [2022]) referred to as DINCAE. Due to computational constraints, DINCAE is applied only to the Chl-a fields of the three datasets. Nevertheless, this comparison provides a robust benchmark against a reference method and allows us to assess whether our approach achieves state-of-the-art reconstruction performance.

Experiment	Architecture	Inputs
SmaAt_phy	SmaAt	SST, clear-sky shortwave radiation, SSH, SSS, 10 m wind vorticity, surface current magnitude, SSI, MLD, Chl-a, PSC (%)
SmaAt	SmaAt	Chl-a, PSC (%)
DINCAE	DINCAE	Chl-a

Table 2. *Summary of the gap-filling experiments and corresponding model inputs. Each architecture is trained independently on each of the three datasets.*

3. Results and Discussion

Table 3 summarizes the performances of each experiment for OC-CCI, GlobC-AVW and GlobC-CMEMS. First, close Chl-a RMSE between DINCAE and SmaAt validates the architecture pertinence for gap-filling. Second, physics added in SmAat_phy help to further improve the network completion on Chl-a. Concerning PSC, the upgrade is less clear.

Metric / Dataset	RMSE Chl-a (mg m^{-3})			IOU PSC		
	AVW	CMEMS	OC-CCI	AVW	CMEMS	OC-CCI
DINCAE	0.403	0.437	0.410	–	–	–
SmaAt	0.396	0.423	0.328	0.733	0.855	0.902
SmaAt_phy	**0.343**	**0.396**	**0.307**	**0.734**	**0.861**	**0.904**

Table 3. *Performance metrics for Chl-a reconstruction (RMSE) and PSC classification (IOU) across datasets for each experiment*

Figure 4 presents the spatial distribution of the normalized reconstruction error for Chl-a. The largest errors are found in coastal regions, as expected, since these areas are less frequently observed by satellites, exhibit stronger variability and concentrations, and are characterized by processes occurring at spatial scales smaller than 1° (Lévy et al. [2012], Mahadevan [2016]). In contrast, oligotrophic regions such as subtropical gyres show the lowest errors, reflecting their relatively weak variability. Errors patterns are broadly consistent across the three datasets, although OC-CCI exhibits higher error values in the open ocean. This behavior is consistent with its higher proportion of missing data, which negatively impacts the reconstruction quality.

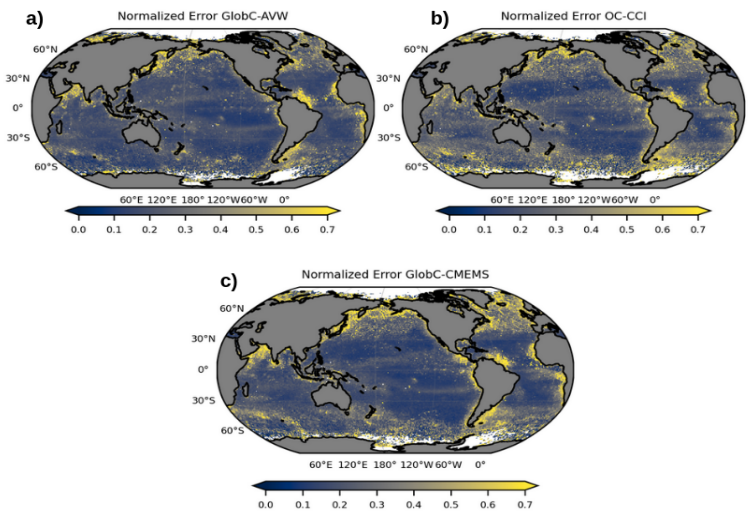

Figure 4. *NRMSE (Eq. 3) of completed daily 1° Chl-a fields on the test set for (a) GlobC-AVW, (b) GlobC-CMEMS, and (c) OC-CCI. NRMSE is computed per grid cell using masked test pixels; highest errors occur in coastal regions, and lowest errors in subtropical gyres, with OC-CCI showing higher open-ocean errors..*

Figure 5 maps the spatial distribution of PSC reconstruction agreement (IoU; Eq. 4) on the test set. Overall, regions of reduced skill coincide with areas of strong small-scale variability and sharp gradients, particularly coastal zones and parts of the equatorial band, where key processes occur at spatial scales smaller than the 1° grid and where satellite sampling is more heterogeneous. These patterns are broadly consistent with the Chl-a NRMSE maps (Fig. 4), which also show the largest errors near coasts and lower errors in relatively stable oligotrophic gyres.

Across datasets, OC-CCI achieves the highest PSC IoU (≈ 0.90), whereas GlobC-CMEMS is lower (≈ 0.86) and GlobC-AVW is substantially lower (≈ 0.73) (Table 3). A simple explanation based on sparsity alone is not sufficient: although GlobC-CMEMS PSC is extremely sparse globally (≈ 93% missing; Table 1), OC-CCI PSC is also highly sparse (≈ 86% missing) yet yields the best skill. A more plausible interpretation is that dataset construction and label consistency differ: OC-CCI PSC are derived using the Brewin et al. [2015] approach, which may produce PSC variability that is more directly aligned

with Chl-a-like optical signals, leading to spatial skill patterns resembling those of Chl-a. In contrast, the PSC products associated with GlobC-AVW (El Hourany et al. [2019]) and GlobC-CMEMS (Xi et al. [2020]) may encode community structure in a way that is less tightly constrained by the information the network receives at 1° daily resolution, making learning and generalization harder, especially under very high missingness. Finally, adding physical predictors yields only marginal PSC improvements (Table 3), suggesting that at this resolution the PSC targets are dominated by product-specific assumptions and/or noise, and that PSC reconstruction may be more limited by target uncertainty and sampling than by a lack of physical covariates.

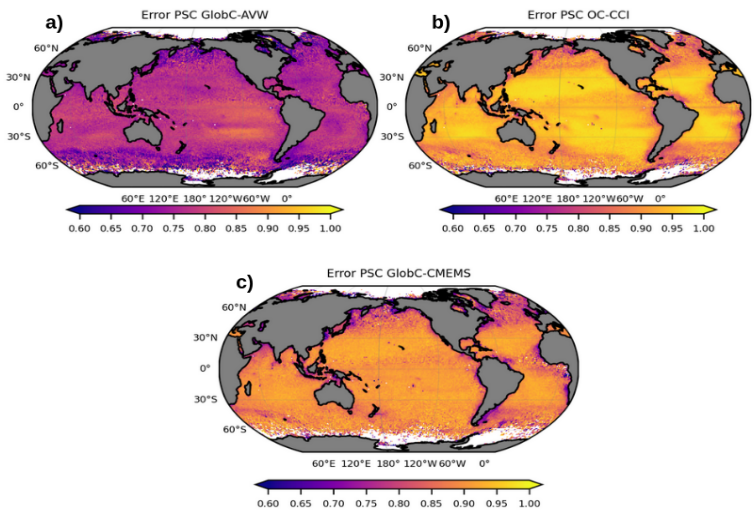

Figure 5. *IoU (Eq. 4) of completed PSC fields on the test set for the three datasets. The soft IoU metric is computed per 1° grid cell on masked test pixels, quantifying agreement between reconstructed and reference PSC relative compositions..*

4. Conclusion

In this study, we investigated the potential of incorporating ocean surface physical fields to improve deep-learning-based gap filling of daily global ocean colour products at coarse spatial resolution. The objective was to produce spatially complete ocean-colour datasets, enabling comparative trend analyses between commonly used products using a consistent gap-filling method. We trained a neural network to complete missing values in three commonly used Chl-a and PSC datasets over the 1998–2023 period at 1° resolution. Physical reanalysis fields from GLORYS12 and ERA5 were included as network inputs to address the strong sparsity of the ocean colour data and take advantage of the relationship between the physical environment and phytoplankton. Our results demonstrate that the inclusion of physical predictors consistently enhances the reconstruction of missing pixels in Chl-a fields. For PSC, however, the benefits of this approach remain to be demonstrated. Further studies could take advantage of recently developed databases, such as those introduced by Xi et al. [2025], and explore alternative groupings of pigment information, for example, by separating pigment-derived phytoplankton communities into functional types rather than size classes. These findings highlight the value of multi-field approaches for the completion of ocean colour products. Beyond improving spatial coverage, such reconstructions provide more consistent datasets for the analysis of long-term ocean colour trends. Ongoing and future work will focus on investigating Chl-a trends using the completed datasets.

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Competing Interests. None

Data Availability Statement. Ocean colour data

- OC-CCI and Brewin PSC: available through the CMEMS platform (OCEANCOLOUR-GLO-BGC-L3-MY-009-107).
- GlobC-AVW: downloaded from the ACRI-ST FTP server (available here)
- Cmems-Globc and Xi PSC: distributed via CMEMS platform (OCEANCOLOUR-GLO-BGC-L3-MY-009-103)
- PSC El Hourany et al. [2019], accessible on demand
- GLORYSV12 E.U. Copernicus Marine Service Information (CMEMS) [2023] (accessible here)
- ERA5 (accessible here), DOI: 10.24381/cds.adbb2d47

CMEMS platform : https://marine.copernicus.eu/fr/acces-donnees
All codes are available at https://github.com/Sma6500/Chl_global_completion

Ethical Standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions.

- Luther Ollier: Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing-orginal draft
- Roy El Hourany: Conceptualization, Data Curation, Validation, Project Administration, Formal Analysis, Methodology, Supervision, Funding, Resources, Writing-orginal draft
- Marina Levy: Conceptualization, Validation, Project Administration, Supervision, Funding, Resources
- Stephane Doléac: Formal Analysis, Methodology, Validation, Formal Analysis, Methodology, Investigation, Software
- Sylvie Thiria: Formal Analysis, Methodology, Validation, Supervision.
- Carlos E. Mejia: Validation, Supervision, Software, Resources, Project Administration.

Supplementary Material. No Supplementary material are provided with the submission.

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